

TP-3: Processing and Document Representation

1. Removing Punctuation

Instruction: Remove punctuation to reduce complexity, as it typically does not hold vital information.

2. Removing URLs

Instruction: URLs should be removed because they do not provide word-level information for text analysis.

3. Removing Stop Words

Instruction: Remove stopwords (common words like articles and prepositions) because they add little to no value to the meaning of the sentence.

4. Lowercasing

Instruction: Lowercase your text to improve general accuracy, as many language models treat capital letters differently. It is generally recommended for an English corpus.

5. Tokenization

Instruction: This is the foundational step in IR-the process of breaking your text (corpus) into individual units called tokens (by Sentence, by Paragraph, or by word).

6. Stemming (Reduction to Root Word)

Instruction: Use stemming to reduce inflected words to their word stem.

Method: It is a heuristic process that cuts off the ends of words.

7. Lemmatization (Reduction to Dictionary Form)

Instruction: Use lemmatization to reduce inflected words to their base or dictionary form, the lemma.

Method: It uses vocabulary and morphological analysis.